# L04 — Speech Recognition

## Phonetics, features, ASR architectures, and WER

Study summary generated from lecture slides. ETH AI for Digital Characters.

### Phonetics

- Phones: speech sounds (ARPAbet/IPA symbols)
- F0 = lowest frequency of periodic waveform; pitch = perceptual correlate
- Mel scale: Mel = 1127*ln(1 + f/700) - perceptual pitch spacing
- Complex waves = sum of sine waves; spectrogram shows frequency over time
- Consonant types: fricatives, stops; vowels have formant structure

### Feature extraction pipeline

- 1. Windowing: 20-40 ms frames, ~50% overlap, Hamming window
- 2. DFT/FFT per frame -> power spectrum P(k)
- 3. Mel filter bank: triangular overlapping filters on Mel scale
- 4. Log mel -> ~80-dimensional feature vector per frame
- Augmentation: time warping, frequency masking, time masking

### Mel filter bank construction

- Convert f_low and f_high to Mel
- Space N+2 points linearly in Mel (for N filters)
- Convert back to Hz; create triangular filters (start-peak-end)

### ASR architectures

- Encoder-decoder with subsampling (CNN pooling, pyramidal RNN)
- CTC: per-frame output + blank token; collapse duplicates; alignment via dynamic programming
- RNN-Transducer: encoder + predictor (LM) + joiner; supports streaming
- Streaming: increment output u on non-blank, acoustic index t on blank
- Whisper: Transformer encoder-decoder, multilingual multitask

### Word Error Rate

- WER = 100 x (Substitutions + Insertions + Deletions) / N_ref
- Lower is better; function-word errors common
- Statistical significance: matched-pair sentence segment word error test